

1. A series RLC circuit $R = 400\Omega$ and $C = 5\text{MF}$. It is connected to an AC source with $f=60\text{Hz}$. the capacitive reactive is

A. 430Ω

B. 530Ω

C. 654Ω

D. 754Ω

2. Kirchhoff's law state

A. the total current flowing away from the junction is equal to the total current flowing towards the junction

B. the algebraic sum of the voltage drops taken around the loop is equal to the result e.m.f. acting in that loop

C. A and B

D. the total current flowing away from the junction is less than the total current flowing towards the junction.

3. What cause ionic polarization?

A. ion splitting

B. cation and anion displacement

C. never occurs

D. magnetic field passing

4. Which of the following statements are correct regarding electromagnetic waves traveling through a vacuum?

A. electric and magnetic fields associated with the waves are perpendicular to each other and to the direction of wave propagation

B. all waves have the same wavelength

C. The speed of the waves depends on their frequency

D. All waves have the same frequency

5. Which of the following pair of quantities are used to describe Heisenberg's Uncertainty principle?

- A. Energy and time
- B. Energy and Angular Momentum
- C. Position and time
- D. Angular position and Energy

6. A coil induces 350Mv when the current changes at the rate of 1A/s. the value of inductance is

- A. 350mH
- B. 250mH
- C. 3500Mh
- D. 150Mh

7. Valence band comprises of electrons that are

- A. freely moving inside the solid
- B. core electrons
- C. Electrons in the outer most orbit of the atom
- D. mobile electrons

8. The term “semiconductor” arises from:

- A. Insulating properties of silicon and GaAS
- B. Resistor-like properties of metal oxides
- C. Variable conductive properties of some materials
- D. The fact that there's nothing better to call silicon

9. Which of the following is most likely to cause sun burn by delivering more energy to individual molecules in skin cells?

- A. Microwaves
- B. Ultraviolet light
- C. Visible light
- D. infrared light

10. A certain light truck can go around a curve having a radius of 150m with a maximum speed of 32.0m/s. To have the same acceleration at what maximum speed can it go around a curve having a radius of 75.0m?

A. 23m/s

B. 32m/s

C. 64m/s

D. 45m/s

11. Which of the following is false about quantization of elastic waves?

A. For elastic mode of angular frequency ω the zero point energy is equal to $1/2\hbar\omega$

B. A phonon is an electromagnetic wave

C. The quantum energy is called a phonon

D. The energy of a lattice vibration is quantized

12. The average energy of a harmonic oscillator in 3-direction is

A. $3kT$

B. None

C. $3/2kT$

D. Kt

13. Which of the following is the β^{-1} particle?

A. ${}_{-1}^0e$

B. ${}_0^1e$

C. ${}_+^0e$

D. ${}_2^4He$

14. Binding energy is

A. The amount of energy released when neutrons change energy levels

B. The amount of energy required to break a nucleus apart into electrons and neutrons

C. The amount of energy required to break a nucleus apart into protons and neutrons

D. The amount of energy required to break a Nucleus apart into Protons and electrons

15. Which formula may be used to compute the energy of a photon?

A. $E=FS$

B. $E=1/2mv^2$

C. $E=mg\Delta h$

D. $E=hf$

16. The chief hazard of radiation is

A. Damage to cells due to heating

B. Damage to living cells due to ionization

C. The creation of new isotopes within the body

D. Damage to living cells due to the creation of chemical impurities

17. What type of laser is used most widely in industrial materials processing applications?

A. YAG laser

B. Dye laser

C. Ruby laser

D. Carbon Dioxide laser

18. A potter's wheel moves uniformly from rest to an angular speed of 1.00 rev/s in 20.0 s. What will be its average angular acceleration in radians per second per second?

A. 0.314

B. 0.414

C. 3.14

D. 4.14

19. Energy of an electron in an atom is

A. Continuous

B. Quantized

C.Radial

D.Randomized

20. Capacitance is the ability of-----to store charge?

A.Battery

B.Capacitor

C.Resistor

D. inductor

21.With reference to waves, what is a trough?

A. The initial position of the wave

B.The final position of a wave

C.The lowermost position of a wave

D. The upper most position of a wave

22.In a p-type material ,electrons are

A.Entirely absent

B. Majority carriers.

C. Minority carriers

D. Positively charged

23. One of the following is not essential conditions for work to be done on an object

A. Force should be perpendicular to displacement

B. Force must be exerted on the object

C.The force should have a component along the line of displacement

D.The force must cause a motion or displacement

24.The equations $\frac{\partial H}{\partial q_i} = -\dot{q}_i$ and $\frac{\partial H}{\partial \dot{q}_i} = q_i$ are called-----of motion

A. Hamilton's equations

B. Newton's equations

C. Time-drift's equations

D. Lagrange's equations

25. If a fixed quantity of an ideal gas is held at a constant volume, which Variable relates to pressure, and what is that relation?

A. Temperature; direct proportionality to pressure

B. Temperature; direct proportionality to square root of pressure

C. Temperature; direct proportionality to square of pressure

D. Temperature; inverse proportionality to pressure

26. What will the absolute pressure at an ocean depth of 1000m. Assume the density of sea water is 1030 kg/m^3 and the air above exerts a pressure of 1013 kPa?

A. $1.02 \times 10^7 \text{ Pa}$

B. $1.02 \times 10^7 \text{ Pa}$

C. $1.02 \times 10^5 \text{ Pa}$

D. $2.02 \times 10^5 \text{ Pa}$

27. Which of the following is not a unit vector?

A. $i + j$

B. $\frac{1}{\sqrt{3}}(i + j + k)$

C. $\sin\theta i + \cos\theta j$

D. $\frac{1}{\sqrt{6}}(i + 2j + k)$

28. If an object of mass m attached to a light spring is replaced by one of mass $9m$, the frequency of the vibrating system change by a factor of

A. $1/9$

B. 9

C. 3

D. $1/3$

29. A quantum of light energy is

A. photo electron

B. Photon

C. Lumen

D. Spectra

30. When the neutron leaves the nucleus its mass

A. decrease

B. Triples

C. increases

D. stays the same

31. The angular momentum of the electron is defined by the quantum number that is denoted as

A.s

B.l

C.n

D.m

32. The nature of binding for a crystal with alternate and evenly spaced positive and negative ions is

A.ionic

B.Inert

C.Covalent

D.Metallic

33. X-rays are used for the study of crystal structure because

A. The crystals are completely transparent to X-rays

B.X-rays are completely absorbed by the crystal

C.The wavelength of X-rays is very small in comparison with the inter- atomic spacing in crystals

C.The wavelength of X-rays is of the same order of the magnitude in the inter-atomic spacing in crystals

34. When resistors with different resistances are connected in series, which of the following must be the same for each resistor?

A. Potential Difference

B. Current

C. Power delivered

D. B and C

35. The product of the area and the fluid speed at all points along a pipe is constant for an incompressible fluid refers to

A. Pascal's law

B. Bernoulli's principle

C. Continuity equation

D. Archimedes' principle

36. A car of mass m traveling at speed v crashes into the rear of a truck of mass $2m$ that is at rest and is in neutral at an intersection. If the collision is perfectly inelastic, what is the speed of the combined car and truck after the collision?

A. $V/2$

B. $2v$

C. V

D. $V/3$

37. In quantum mechanics, the operator describing kinetic energy is represented by:

A. $i\hbar \frac{\partial}{\partial x}$

B. $\frac{-\hbar^2 \partial^2}{2m \partial x^2}$

C. $-i\hbar \frac{\partial}{\partial x}$

D. $i\hbar \frac{\partial}{\partial x}$

38. The relation between the number of microstates $\Omega(E)$ for an ideal gas of N monoatomic molecules in a volume V is related to energy E by: (Planck).

A. $\Omega(E) \propto E^{\frac{2N}{2}}$

B. $\Omega(E) \propto E^N$

C. $\Omega(E) \propto E^{\frac{N}{2}}$

D. $\Omega(E) \propto E$

39. DC current produces a/an

A. Electric field

B. Magnetic Field

C. Gravitational field

D. Electromagnetic field

40. Suppose 6 white and 6 red balls are placed in a box. In the process of randomly drawing one ball at a time, what is the probability of drawing two white balls successively?

A. 1/12

B. 5/11

C. 1/2

D. 1/6

41. Three capacitors each of the capacitance C are given. The resultant capacitance $\frac{2}{3}C$ can be obtained by using them

A. All in series

B. Two in parallel and third in series with this combination

C. Two in series and third in parallel across this combination

D. All in parallel

42. Hamilton's equations are also called-----

A. Legendre's equations

B. Canonical equations

C. Linear equations

D. None of these

43. Which of the following is true about laminar flow of fluid?

A. The answer is not given

C. The velocity of the fluid particles at any point is changes with time

C. The velocity of the fluid changed only at some points

D. The velocity of the fluid at any given point does not change

44. Which one of the following set of parameters contains only intensive parameters?

A. Temperature, pressure and volume

B. Density, pressure and heat capacity

C. Temperature, density and specific heat

D. Pressure, temperature and internal energy

45. An object-spring system moving with simple harmonic motion has an amplitude A . When the kinetic energy of the object equals the potential energy stored in the spring, what is the position x of the object?

A. A

B. $\frac{A}{\sqrt{2}}$

C. 0

D. $\frac{A}{\sqrt{3}}$

46. The uncertainty principle states that the error in measurement is due to-

A. Due to the small size of particles

B. Dual nature of particles

C. Due to large size of particles

D. Due to error in measuring instrument

47. Two cylinders A and B at the same temperature contain the same quantity of the same kind of gas. Cylinder A has three times the volume of cylinder B. What can you conclude about the pressures the gases exert?

- A. Pressures must be equal
- B. The pressure in A must be one-third the pressure B.
- C. The pressure in A is three times the pressure in B
- D. We can conclude nothing about the pressures

48. The number of degrees of freedom is equal to the number of _____

- A. Particles
- B. Generalized Coordinates
- C. Constraints
- D. A & C

49. Probability is the most common term in quantum mechanics. This is mainly due to;

- A. uncertainty principle
- B. Quantization of energy
- C. Macroscopic size of particles treated IN Quantum Mechanics
- D. A & B

50. Which of the following property is not a thermodynamic property of the system?

- A. Heat transferred during the cycle
- B. Difference of the work and heat
- C. Work done during the cycle
- D. Total entropy generated

